StressFold: Perturbation-response profiling for tabular model generalization

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Abstract

Overfitting is often reduced to a train-test gap, while robustness is reported as a second accuracy number and synthetic data are treated as additional evidence. Those quantities answer different questions. StressFold is a model-agnostic protocol for tabular classification and regression that keeps them separate. It estimates clean generalization under an explicit split policy, traces paired perturbation-response curves under declared measurement or distribution shifts, and runs falsification tests that repeat the complete learning workflow. Synthetic replicas are admitted only as audited stress instruments; they do not enlarge the effective sample size or substitute for real holdout observations. We specify estimands, leakage boundaries, perturbation operators, paired Monte Carlo aggregation, and generator diagnostics. Controlled experiments show three failure modes that a single score conceals: an interpolating tree with poor population loss can improve under randomized measurement noise; common perturbation draws materially reduce comparison variance; and replicas with accurate one-dimensional marginals can destroy class-conditional dependence and downstream utility. The protocol is implemented as a deterministic, local-first experiment specification and produces report-ready evidence rather than an "overfit probability."

Keywords: generalization, robustness, permutation tests, synthetic data, tabular learning, Monte Carlo.

1 Technical summary

StressFold is designed around a negative claim: no finite collection of resamples or generated rows can certify that a fitted model will generalize to every future environment. What can be measured is narrower and useful. Given a complete learning algorithm, a split policy, a loss, and a family of perturbations, the protocol produces an evidence profile with three non-interchangeable components:

1. **Generalization** estimates predictive risk on observations excluded from all fitting and selection.
2. **Robustness** estimates the change in risk under a named perturbation mechanism and severity scale.
3. **Falsification** asks whether the entire observed workflow could produce its result under a specified null.

The distinctions are operational. Cross-validation estimates the performance of an algorithm trained on comparable unseen samples, not necessarily the conditional risk of the single final fit; fold-level standard errors can also be much too small (Bates et al., 2024). Common-corruption benchmarks show why response curves are more informative than a single shifted test set (Hendrycks and Dietterich, 2019), while natural-domain benchmarks show that convenient synthetic shifts are not replacements for real changes across time, sites, devices, or populations (Koh et al., 2021).

The output is therefore a set of losses, paired differences, intervals, and explicit assumptions. It is not a composite traffic light. A model may generalize well on clean data yet be sensitive to measurement error; a highly adaptive model may fit random labels yet still exploit genuine structure (Zhang et al., 2017); a generator may match marginals while omitting predictive dependence. Combining these outcomes would erase the information the audit is meant to expose.

2 Clean fit and perturbation response can disagree

Figure 1 gives the central controlled example. The data-generating process (DGP) is known, so population Brier loss and label-preserving measurement error can be evaluated without assuming that arbitrary noisy rows retain their labels. Increasing tree depth drives training loss to zero but raises population loss after depth two. Under perturbation, the depth-three model steadily loses skill. The interpolating tree instead improves: random input displacement moves some extreme, sample-specific leaf probabilities toward a useful randomized decision. The improvement is real for this perturbation distribution, but it does not rescue the tree's poor clean population loss.

This example also fixes the interpretation of a response curve. It is conditional on an operator, its units, its support constraints, and a label semantics. A curve can diagnose brittleness or accidental smoothing. It cannot identify the deployment distribution unless the operator is itself a credible model of that distribution.

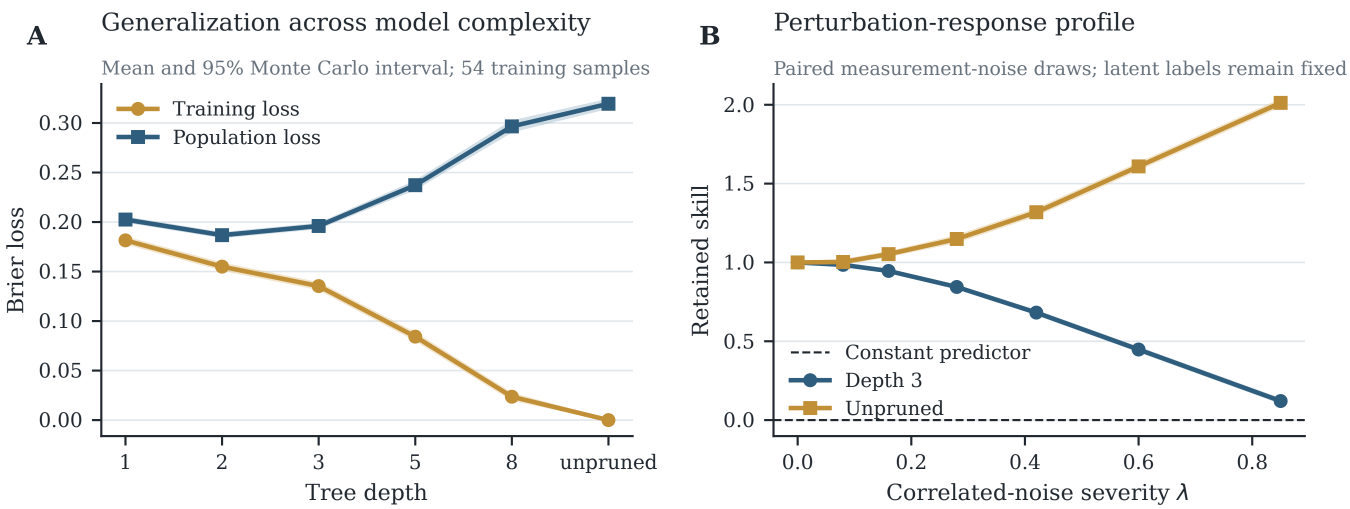

Figure 1: Generalization and robustness are distinct estimands. (A) Mean Brier losses and 95% Monte Carlo intervals across 54 independent training samples of size 320; population loss uses 2,600 fresh cases per sample and the known Bernoulli probabilities. (B) Retained skill under correlated measurement noise for a fixed sample of 420 training and 3,200 latent audit cases, averaged over 44 common perturbation draws. Values above one indicate improvement relative to clean skill, not super-generalization. The interpolating tree's noise response illustrates why robustness cannot be used as a proxy for clean predictive risk.

3 Audit object and estimands

3.1 The complete algorithm is the unit under test

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be sampled from population P. The audited algorithm A maps a training sample to a predictor,

$$f_{\mathcal{T}} = A(\mathcal{T}).$$

Here A includes imputation, encoding, scaling, feature selection, hyperparameter search, probability calibration, threshold selection, and fitting. Treating only the final estimator as A leaves selection leakage outside the audit.

An outer split policy produces $(\mathcal{T}_b, \mathcal{H}_b)$, $b = 1, \ldots, B$. The policy is part of the scientific specification: stratified random splitting is appropriate only for exchangeable rows; repeated entities require grouped splits; temporal prediction requires rolling-origin splits; dependent sequences require blocks. Preprocessing statistics, perturbation calibration, and synthetic generators are fit using $\mathcal{T}_b$ only.

3.2 Generalization

For a fitted predictor $f_{\mathcal{T}}$, population risk under loss ℓ is

$$\mathcal{R}_P(f_{\mathcal{T}}) = \mathbb{E}_{(X,Y)\sim P}[\ell(f_{\mathcal{T}}(X), Y) \mid \mathcal{T}]. \tag{1}$$

The clean audit estimate is $\widehat{\mathcal{R}}_{b,0} = |\mathcal{H}_b|^{-1} \sum_{i\in\mathcal{H}_b} \ell(f_{\mathcal{T}_b}(x_i), y_i)$. The descriptive generalization gap

$$G_b = \widehat{\mathcal{R}}_{b,0} - |\mathcal{T}_b|^{-1} \sum_{i\in\mathcal{T}_b} \ell(f_{\mathcal{T}_b}(x_i), y_i) \tag{2}$$

is useful but not sufficient: a small gap can accompany uniformly poor predictions, and a flexible model can generalize despite a near-zero training loss. For probabilistic classification the default loss is log loss or Brier loss; for regression it is MAE or RMSE. Accuracy, balanced accuracy, AUROC, subgroup error, and calibration are secondary views rather than substitutes for a proper primary loss.

If tuning is performed, outer assessment surrounds the complete inner selection. Define selection optimism on the loss scale as

$$O_b = \widehat{\mathcal{R}}_{b,\text{outer}} - \widehat{\mathcal{R}}_{b,\text{inner selected}}. \tag{3}$$

Positive O_b means the selected configuration looked better during search than it performed outside that search. A locked holdout supports uncertainty conditional on one fitted workflow. Nested cross-validation supports algorithm-level assessment when a single holdout would be too costly. These targets should be named rather than blended (Bates et al., 2024).

3.3 Robustness

Let $Q_{\lambda,r}$ be perturbation draw r at severity λ, calibrated on $\mathcal{T}_b$. It maps an audit case to $(X^{\lambda,r}, Y^{\lambda,r})$. The perturbation risk is

$$\mathcal{R}_{P,Q_\lambda}(f) = \mathbb{E}_P \mathbb{E}_{Q_\lambda}\left[\ell(f(X^\lambda), Y^\lambda)\right], \tag{4}$$

and the excess response is $\Delta(\lambda) = \mathcal{R}_{P,Q_\lambda}(f) - \mathcal{R}_{P,Q_0}(f)$. For a constant-predictor reference L_{ref}, retained skill is

$$S(\lambda) = \frac{L_{\text{ref}}(\lambda) - \mathcal{R}_{P,Q_\lambda}(f)}{L_{\text{ref}}(0) - \mathcal{R}_{P,Q_0}(f)}. \tag{5}$$

StressFold reports the full $S(\lambda)$ curve, a prespecified degradation area, and the first severity at which $S(\lambda) \leq 1/2$, when crossed. Areas are never pooled across operators whose severity scales have different meanings.

Two label semantics are allowed. A *measurement perturbation* changes the representation while holding the latent case and label fixed. A *semantic perturbation* changes the case itself and requires a label oracle or known simulator. If neither claim is defensible, the protocol reports prediction movement or class-flip probability, not accuracy against a retained label.

3.4 Falsification

For exchangeability blocks C, the permutation null is

$$H_0 : Y \perp X \mid C.$$

Let $T(A, \mathcal{D})$ be a higher-is-better statistic produced by the complete pipeline. With M valid label permutations,

$$p = \frac{1 + \sum_{m=1}^{M} \mathbb{I}\{T(A, \mathcal{D}_m^{\pi}) \geq T(A, \mathcal{D})\}}{M + 1}. \tag{6}$$

Every permutation repeats preprocessing, selection, and tuning. This test asks whether the workflow found structure beyond the stated null (Ojala and Garriga, 2010); it does not prove that the structure is causal, non-leaking, stable, or useful in a new domain. A separate random-label capacity curve corrupts a fraction of training labels, refits, and contrasts corrupted-label training fit with clean audit loss. That curve describes memorization capacity and carries no significance interpretation.

4 Perturbation operators

Every operator has a training-only calibration map, a severity scale with units, a constraint policy, and a declared label semantics. Defaults favor empirical support and reversible assumptions.

The covariate tilt avoids inventing impossible feature combinations. With

$$w_i(\delta, v) = \frac{\exp(\delta v^\top z_i)}{\sum_j \exp(\delta v^\top z_j)}, \qquad n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2},$$

the direction v can be a domain feature, a subgroup contrast, or a principal component. A small n_{eff} warns that a nominally smooth shift is supported by few rows. Real site, time, or device partitions supersede this synthetic operator whenever available (Koh et al., 2021).

5 Paired protocol and aggregation

For each outer split, StressFold uses common audit cases and common random perturbations across candidate models. For model pair (a, c), split b, severity λ, and draw r, define

$$D_{b\lambda r} = \widehat{\mathcal{R}}_{b\lambda r}(f_a) - \widehat{\mathcal{R}}_{b\lambda r}(f_c). \tag{7}$$

The split-level contrast is $\bar{D}_{b\lambda} = R^{-1} \sum_r D_{b\lambda r}$. Aggregation occurs over draws within split before summarizing over outer splits. Perturbed replicas of one audit row are not independent new observations, and the nominal sample size is never multiplied by R.

The practical protocol is:

1. Validate schema, constraints, row identity, grouping, and temporal order; scan exact and near duplicates across the intended split.

2. Construct outer splits. Inside each $\mathcal{T}_b$, fit all transforms and select hyperparameters. Freeze f_b.

3. Evaluate clean $\mathcal{H}_b$, then apply training-calibrated operators to that same audit fold at a prespecified severity grid and seed schedule.

4. Refit A on cluster- or block-bootstrap versions of $\mathcal{T}_b$ and predict the unchanged $\mathcal{H}_b$ to measure refit instability.

5. Run label permutations within valid blocks, repeating the complete workflow. Run training-label corruption as a separate descriptive experiment.

6. Fit any replica generator on $\mathcal{T}_b$ only; audit it against real $\mathcal{H}_b$ before using its rows as stress probes.

7. Export losses, paired contrasts, uncertainty, constraints, split indices, seeds, and software versions. Do not emit a single global score.

For a locked holdout, uncertainty in an additive loss difference is obtained by paired bootstrap resampling of audit cases after averaging their perturbation draws. For grouped or temporal observations, resampling respects clusters or blocks. If repeated cross-validation is used, ordinary fold standard errors are not reported; fold overlap induces dependence, and dedicated nested-CV intervals are preferred (Bates et al., 2024). When many severities are used for formal testing, a prespecified maximum statistic or multiplicity correction replaces a row of unadjusted p-values. Response bands are descriptive by default.

6 Controlled checks of aggregation and falsification

Figure 2 isolates two protocol choices. First, paired perturbations give each model the same Monte Carlo difficulty. At $R = 16$, independent draws had 1.83 times the estimator standard deviation of common draws in the deterministic experiment. Pairing does not remove sample

Table 1: Core operators. D contains training-fold robust scales and $LL^\top$ is a shrinkage correlation estimate. Operators are run separately so that each curve retains an interpretable horizontal axis.

Operator	Mechanism	Valid interpretation
Correlated numeric noise	$x'_{\mathrm{num}} = x_{\mathrm{num}} + \lambda DL\epsilon$, $\epsilon \sim N(0, I)$; clip only to declared physical bounds and report clipping.	Measurement robustness when the latent label is invariant to sensor error.
Categorical transition	Independently retain level a, or sample $a' \sim K_j(\cdot \mid a)$, with transition mass controlled by λ.	Coding or observation error when transitions are plausible; otherwise prediction stability only.
Missingness	Apply a mask at rate λ. MCAR is explicit; MAR requires a user-supplied mechanism.	Robustness to the named observation process, including the fitted imputer.
Tail contamination	Replace a prespecified fraction of cells or rows with empirical-tail values subject to schema constraints.	Sensitivity to outliers within observed support, not adversarial robustness.
Covariate tilt	$w_i(\delta, v) \propto \exp(\delta v^\top z_i)$; resample complete observed rows and report effective sample size.	Reweighted empirical shift along a named risk direction; observed labels remain attached.
Training-label corruption	Replace a fraction q of training labels and rerun A; never corrupt the clean audit labels.	Memorization and regularization response, not a deployment-performance estimate.
Local smoothed bootstrap	Resample a complete row, perturb numeric fields locally, and modify categories within a declared neighborhood.	Local support probing; labels are retained only for features declared label-invariant.

uncertainty, but it avoids paying unnecessary simulation variance in a model comparison.

Second, a permutation test must repeat selection. Under a pure null, 40 random binary predictors were scored on the same 160 cases and the best was selected. Permuting labels while holding that winner fixed produced a 72.1% false-positive rate at nominal 5%. Repeating the full 40-way selection inside each of 199 permutations reduced the empirical rate to 5.7% over 140 null repetitions. The example is deliberately stark: it makes the conditioning error visible rather than relying on asymptotics.

Permutation evidence should remain modestly phrased. A low p-value says that the workflow's statistic is unusual under the encoded exchangeability null. A leaked target surrogate is also unusual under that null. Duplicate detection, group-aware splitting, time-aware splitting, and domain review are therefore complementary controls, not optional preprocessing.

7 Synthetic replicas are audited instruments

A learned generator $G(\mathcal{T}_b)$ is a random function of the same finite training sample. Its rows can expose local instability, probe constraints, or exercise software. They do not provide new information about unobserved modes of P, and they cannot enlarge the effective sample size of a confidence interval. A diffusion generator such as TabDDPM can model heterogeneous tables (Kotelnikov et al., 2023), but model complexity does not alter this information boundary.

StressFold therefore evaluates a generator along separate axes:

- **Fidelity:** held-out classifier two-sample testing (C2ST) (Lopez-Paz and Oquab, 2017) and maximum mean discrepancy (MMD) with permutation calibration (Gretton et al., 2012);
- **Coverage:** marginal, pairwise, subgroup, and target-conditional discrepancies, with precision/recall-style support diagnostics (Alaa et al., 2022);
- **Authenticity:** exact copies and train-to-synthetic versus holdout-to-synthetic nearest-neighbor structure;
- **Utility:** a fixed reference model trained on synthetic data and evaluated on the untouched real holdout (TSTR).

C2ST AUC near one indicates a detectable difference; a value near one half is a failure to distinguish, not proof of equality. MMD has the same power caveat. Synthetic-only evaluation is not treated as unbiased real-population evaluation; methods that combine generated and real labels require an explicit correction design (Boyeau et al., 2025).

Figure 3 constructs the failure analytically. Draw $Y \sim$ Bernoulli(1/2), $S \in \{-1, 1\}$ uniformly, and

$$X_1 = S+\epsilon_1, \qquad X_2 = (2Y-1)S+\epsilon_2, \qquad \epsilon_j \sim N(0, 0.3^2).$$

Each class has identical one-dimensional marginals but opposite dependence. Training contains 850 rows and the real holdout 3,200. The empirical bootstrap copies all generated rows, the smoothed bootstrap adds $N(0, 0.12^2)$ noise, and the marginal generator samples X_1 and X_2 independently within class. Table 2 shows why generator

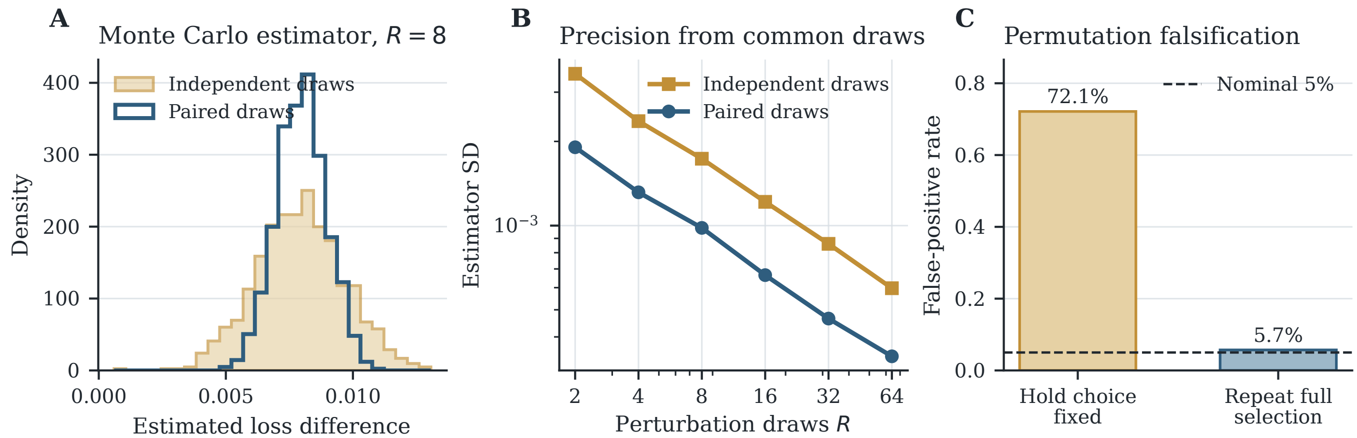

Figure 2: Pairing improves precision; full-workflow permutation preserves the null. (A) Sampling distributions of the depth-three minus depth-four loss contrast at severity $\lambda = 0.42$, using eight draws and a pool of 1,200 deterministic perturbations. (B) Estimator standard deviation over 900 Monte Carlo resamples at each R. (C) Null false-positive rates for a selected random predictor when model choice is held fixed versus repeated inside every permutation.

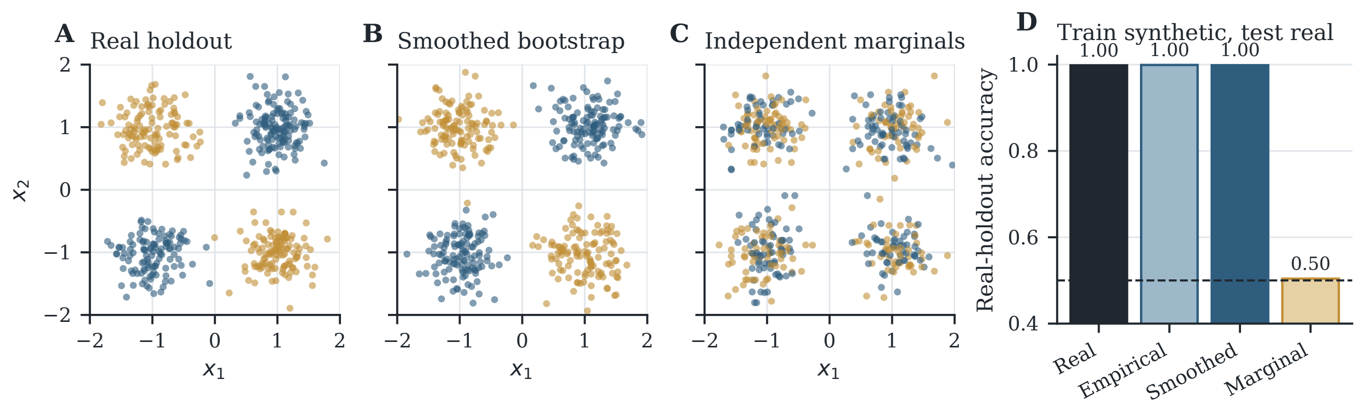

Figure 3: Marginal fidelity does not imply predictive fidelity. In the known DGP, class labels select opposite diagonal modes. A smoothed row bootstrap preserves that joint structure, while independently sampling each feature within class preserves conditional one-dimensional marginals but destroys dependence. The marginal generator's mean conditional KS discrepancy is only 0.059, yet its real-holdout TSTR accuracy is 0.504. Bar labels are rounded; the dashed line is chance.

claims require more than one diagnostic. The orientation-free C2ST value is $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$.

The empirical bootstrap has excellent utility and dependence fidelity but a 100% exact-copy rate. The smoothed bootstrap removes exact copies in this continuous example while retaining utility. Neither result is a privacy guarantee. The marginal generator appears acceptable under univariate checks yet fails both dependence and downstream utility. A strong audit keeps those conclusions separate rather than averaging them into a generator score.

8 Reproducibility and implementation boundary

The reference experiment script, `scripts/reproduce_paper.py`, is independent of the package API. It uses NumPy, SciPy, pandas-compatible CSV output, Matplotlib, and scikit-learn with a fixed root seed of 20260729 and single-threaded estimators. It regenerates every plotted PDF/PNG and both LaTeX/CSV tables. The DGP equations, sample sizes, replicate counts, model settings, and Monte Carlo schedules are encoded directly in that script.

The current interfaces are bounded implementations of this reference protocol, not interchangeable embodiments of every equation above. Browser protocol 0.3 uses independent standardized numeric noise, a protected retained-skill denominator, and a tie-adjusted descriptive null rank. Python package 0.1 reports a pooled paired null-exceedance rate across repeated holdouts and does not call it a permutation p-value. Equation (6) requires coherent dataset-level permutations, a complete workflow rerun, and one prespecified aggregate statistic per permutation.

A production run should serialize more than an image: data and schema fingerprints; target and feature

Table 2: Synthetic-replica diagnostics. KS is the mean class-conditional feature statistic; correlation error is the mean absolute class-conditional correlation difference. C2ST compares the joint (X, Y) distribution. TSTR is real-holdout accuracy.

Replica	KS ↓	Corr. err. ↓	C2ST AUC ↓	TSTR ↑	Copies ↓
Real training data	0.000	0.000	0.500	0.998	0.0%
Empirical bootstrap	0.069	0.004	0.579	0.999	100.0%
Smoothed bootstrap	0.044	0.012	0.535	0.998	0.0%
Independent marginals	0.059	0.916	0.780	0.504	0.5%

roles; constraints; exact split indices; estimator and preprocessing specifications; seed tree; perturbation parameters; rejected-row counts; software versions; and raw per-case/per-draw losses. Report generation is downstream of this immutable run record. A local-first browser interface can compute a bounded baseline directly, as the current lab does, or orchestrate a richer local Python audit. Deep generators are optional extras rather than browser dependencies.

The controlled results are deterministic conditional on the documented software stack, subject to small floating-point and library-version differences. The simulation intervals quantify repetition in the stated DGP, not epistemic uncertainty about whether that DGP resembles a user's deployment environment.

9 Limitations and open questions

Operator validity is the main limitation. Gaussian noise, empirical tilts, and local resampling are convenient, not universal. A response curve is only as relevant as its mechanism. Natural held-out domains should be used whenever available, and failure to observe a natural shift must not be translated into evidence of robustness.

Labels may cease to be valid. Large feature perturbations can move a row across the true decision boundary. Without a simulator or label oracle, retained-label accuracy then mixes model error with annotation invalidity. StressFold falls back to prediction consistency and reports the absence of a correctness claim.

Exchangeability can fail. Permutation tests, ordinary bootstrap, and random splitting all rely on a sampling unit. Households, patients, devices, sites, and time intervals require explicit blocks. A valid blocked null may have few permutations and low power.

Generator diagnostics are incomplete. C2ST and MMD can miss differences in high dimensions or small samples; nearest-neighbor copying is not a privacy proof; TSTR depends on the reference model. The fidelity, coverage, authenticity, and utility dimensions proposed for generative-model auditing (Alaa et al., 2022) are complementary rather than exhaustive.

The protocol detects risk signals, not causes. A large train-audit gap can arise from variance, shift, leakage, or an inappropriate split. A steep perturbation curve can reflect a genuinely hard measurement problem. The evidence profile narrows hypotheses; domain investigation determines the remedy.

The most valuable extensions are empirical rather than ornamental: block-aware operators for longitudinal data; user-defined invariance contracts; simultaneous confidence bands with explicit multiplicity targets; and benchmark suites with predeclared DGPs, leakage traps, and expected operating characteristics.

10 Conclusion

StressFold turns an imprecise question–"is this model overfit?"–into auditable questions with different estimands. Clean outer assessment measures predictive generalization under a split policy. Perturbation-response curves measure sensitivity to named changes. Full-workflow permutation and random-label experiments falsify specific explanations. Synthetic replicas supply controlled probes only after their fidelity, dependence, copying, and utility have been audited.

The controlled experiments show why the separation matters: clean population loss and noise response can move in opposite directions; paired Monte Carlo makes comparisons more precise; incomplete permutation invalidates inference; and good marginals can coexist with useless synthetic dependence. A restrained report of these facts is more scientifically useful than a universal score, and more reproducible than an opaque generator demo.

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